

Permanent Branches

main

- Stable version
- Adviser demo version
- Deployment version

develop

- Active development branch
- All completed features merge here first via Pull Requests

Sprint 1: Baseline Architecture & Auth

July 8 – July 15 At the start of Sprint 1, each developer creates their branch from develop.

Jefferson (Lead Full-Stack / Backend)

Create:

```
git checkout develop
```

```
git pull origin develop
```

```
git checkout -b feature/project-setup
```

Tasks:

- Initialize root Django REST Framework repository architecture
- Create custom `tbl_user` model with role configurations
- Establish database rules, indices, and constraints inside PostgreSQL

Merge: `feature/project-setup` → `develop`

Veejay (Full-Stack / Backend)

Create:

`git checkout develop`

`git checkout -b feature/jwt-authentication`

Tasks:

Implement `/api/v1/auth/register/` and `/api/v1/auth/login/` endpoints

Integrate secure JWT Access & Refresh token mechanics

Set up endpoint route protection middleware

Keith (Frontend Mobile - React Native)

Create:

`git checkout develop`

`git checkout -b feature/mobile-auth-ui`

Tasks:

- Build Student Mobile Login screen layout
- Build Student Mobile Registration screen layout
- Integrate JWT local storage logic on the device

Merge: `feature/mobile-auth-ui → develop`

Zaki (Frontend Web - ReactJS)

Create:

`git checkout develop`

`git checkout -b feature/web-auth-ui`

Tasks:

- **Build OSA Admin Web Login view layout**
- **Scaffold foundational responsive dashboard shell and navigation grid**
- **Integrate Axios configuration mapping login requests to backend token routes**

Merge: feature/web-auth-ui → develop

Sprint 2: Core Intelligent Agent & Ticket Automation Engine

July 16 – July 26

Jefferson (Lead Full-Stack / Backend)

Create:

git checkout develop

git checkout -b feature/openai-intent-classification

asks:

- **Bind external OpenAI API endpoints to backend text routing logic**
- **Construct strict prompt engineering system instructions prioritizing Taglish text validation**
- **Build content matching logic pairing inquiries against the tbl_faqs matrix**

Merge: feature/openai-intent-classification → develop

Veejay (Full-Stack / Backend)

Create:

git checkout develop

git checkout -b feature/ticket-escalation-logic

Tasks:

- Establish schemas and operational routes tracking entries inside `tbl_inquiry_logs`
- Program trigger handlers to instantiate records inside `tbl_ticket` upon fallback detection
- Develop administrative updating routes to claim (`resolved_by`) and update status states

Merge: feature/ticket-escalation-logic → develop

Keith (Frontend Mobile - React Native)

Create:

git checkout develop

git checkout -b feature/mobile-chat-screen

Tasks:

- Program real-time message stream handlers inside the student dialogue container
- Build text message components distinguishing AI responses from student input blocks
- Design the Ticket History layout allowing students to monitor open support status parameters

Merge: feature/mobile-chat-screen → develop

Zaki (Frontend Web - ReactJS)

Create

git checkout develop

git checkout -b feature/web-ticket-board

Tasks:

- Construct the central Web Ticket Grid rendering real-time incoming issues

- **Build the Admin FAQ Management page supporting creation, updates, and removal of database records**
- **Integrate status badge color indicators tracking individual ticket lifetimes**

Merge: feature/web-ticket-board → develop

Sprint 3: Analytics, System Evaluation, and Deployment

July 27 – August 6

Jefferson (Lead Full-Stack / Backend)

Create:

git checkout develop

git checkout -b feature/deployment-config

Tasks:

- **Deploy Django REST Framework API live to a DigitalOcean cloud environment**
- **Configure production database migration parameters targeting live PostgreSQL**
- **Secure structural production secrets using environment parameters**

Merge: feature/deployment-config → develop

Veejay (Full-Stack / Backend)

Create:

git checkout develop

git checkout -b feature/analytics-aggregation

Tasks:

- **Develop system analytics reporting endpoints providing volume, frequency, and time metrics**
- **Conduct comprehensive backend validation and end-to-end route stress testing**

Merge: feature/analytics-aggregation → develop

Keith (Frontend Mobile - React Native)

Create:

git checkout develop

git checkout -b feature/mobile-sus-survey-link

Tasks:

- **Integrate a redirection handler linking students to the external Google Forms evaluation tool**
- **Refine mobile navigation workflows, padding layouts, and alignment constraints**

Merge: feature/mobile-sus-survey-link → develop

Zaki (Frontend Web - ReactJS)

Create:

git checkout develop

git checkout -b feature/web-analytics-dashboard

Tasks:

- **Design and integrate visual metrics charts detailing inquiry density maps**
- **Build data filtration inputs isolating tracking logs by user keys, dates, and intent categories**
- **Deploy the production-built ReactJS dashboard bundle**

Merge: feature/web-analytics-dashboard → develop

End of Every Sprint Workflow

All completed feature branches must go through this exact merging routine:

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All completed feature branches must go through this exact merging routine:

1. Merge completed code to local develop and push:

git checkout develop

git merge feature/your-feature

git push origin develop

2. Submit a Pull Request on GitHub matching feature/* to develop.

3. At the completion of a full sprint cycle, shift the staging branch into deployment status:

git checkout main

git merge develop

git push origin main

